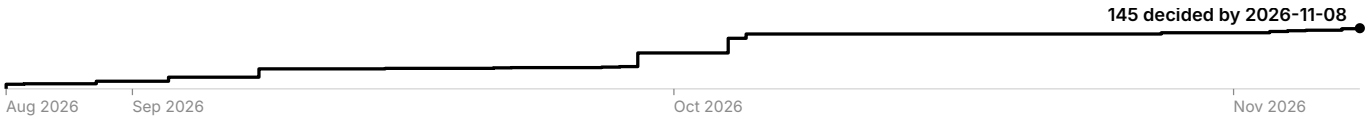
Carlos Toledo

Independent research engineer
exact verification of machine-generated claims · benchmark and grader audits · verifiers that cannot be gamed

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I check what AI systems and benchmarks claim — in exact arithmetic, with code that shares nothing with the claimant — and publish each verdict as a re-runnable certificate: certified, refuted with a witness, or refused with the reason.

145	10	3,582 / 4,000	9.75×10^{15}
CLAIMS DECIDED	ASTERISKS REMOVED	WRONG ANSWERS A TOLERANCE GRADER ADMITS	PAIRS OF POINTS DECIDED EXACTLY
since August 2026, each with the record that decided it	from Tao's optimization-constants registry, on our replays	a certificate grader admits none	the Qiushi Engine's nineteen kissing bounds



The register (certs/claims-ledger.json), read when this page was built: decided claims by day; a solid step, because every count is exact.

BENCHMARK AND RELEASE AUDITS — 2026 · all public

OpenAI's math release — 719 machine-generated manuscripts and 416 Lean challenges, audited at a pinned commit · report

- Every proof re-run under a second, independent kernel (nanoda, written in Rust): 407 of 416 accepted by both kernels and none rejected; the other 9 exceeded the runners' memory or time.
- Every formal statement read against the claim it carries: 34 of 242 formalized families state a different or weaker theorem than their headline, with nothing saying so; 14 gaps published, each pinned to file and line; a kernel fix posted upstream.

Two claimed refutations of the Exponential Time Hypothesis (OpenAI, Anthropic), replayed · report

- Both proofs accepted by two kernels; the statements read against the textbook ETH by two independent readers; the textbook ETH written once in Lean and each lab's statement proved to refute it in its own machine model.
- Both chains then carried to the standard multitape machine in one Lean each; two results merged into complexitylib.

The graders of mathematical benchmarks, measured against 4,000 provably wrong answers · report

- Absolute-tolerance rules accept 3,582 of them; a certificate grader accepts none and rejects nothing right.

METR's time-horizon estimator, re-derived exactly · report

- 44 of 44 fits certified over METR's public runs; the post-2023 doubling time re-derives to between 128.740 and 128.741 days, against the printed 128.744.

GSM8K's answer key, every arithmetic step re-decided · report

- 8,792 keys (test and train); 4,282 annotated steps, all exact; 26 printed steps wrong, 7 of them leaving the key's answer underived.

INDEPENDENT CERTIFICATION OF MACHINE-GENERATED MATHEMATICS — 2026

The Qiushi Engine's kissing-number bounds · report

- 19 of 19 announced bounds witnessed and every pair of 9.75×10^{15} decided exactly; the claimant published its point sets at the audit's request and credits the check in its README. Also the first independent certification of $K(18) \geq 8{,}358$ (34,923,903 pairs, 0 violations).

Registries of AI-generated certificates: Terence Tao's optimization-constants registry removed 10 asterisks on our exact replays (four merged pull requests) and cites [T2026] (report); its PR queue 9 certified, 1 refused (report); AlphaEvolve's community certificates 70 certified, 2 refused (report); Written on the Wall II 5 of 5; C_{89}'s author thanked the replays.

Erdős problems against DeepMind's statements: #488, Gessel's GPT-6 Astra refutation certified by two kernels and two deciders, all 4 planted fakes caught (report); #174, Pálvölgyi's Theorem A.3 proved in Lean as stated, on Palomar (report).

"ℤ is Diophantine in ℚ" (Glazer, AI-produced): the 292,890,231,905-variable count re-derived; the Version 4 quartic's 551 certificate steps replayed; the Lean target proved from eight named hypotheses (report).

Also: OpenAI's Navier–Stokes blowup, a clause missing at the announced commit (report); Bandeira's 0.2(a), two gaps named (report); λ(4), re-certified by a third party (report); $\delta_3 = 117/2192$.

FINANCIAL MODEL RE-PERFORMANCE — October 2026

PU Verificado — every Tesouro Direto price since 2004 re-decided from the Treasury's own rule table · report

- 145,679 prices (2004–2026): 132,538 reproduced to the centavo, 12,791 consistent with the printed rate's precision, 289 off with the exact value and the difference, 61 needing data the publisher kept; two deciders.

RL ENVIRONMENTS, VERIFIERS AND THE VISUAL GRAMMAR — 2026

Three reward designs, one environment, on Prime Intellect's Hosted Training — a pre-registered GRPO study: an answer key and an exact grader both extinguished abstention within ~10 steps, a ternary grader made the model abstain on everything. On the Environments Hub: `lattice-claims`, `blind-spot`, `break-the-grader`.

WARRANT, a visual grammar in which every mark says what it rests on · the rules

- The stroke says how a number was made, words carry the verdict; the site refuses to build a page that breaks a rule.

Writing: cert-machine: independent exact certification of machine-generated mathematics (method paper, draft, not peer-reviewed) · PDF. Outside re-derivations of the verdicts, all concordant, are on the record with their authors' names (report).

HOW I WORK

Verifier first: a check that has never gone red is decorative, so every battery carries planted forgeries. Pre-registration before any spend, deviations dated, negative results reported; every number read from a record, this page's included.

Research: Python (standard library, Inspect, verifiers, pytest), JavaScript (BigInt exact arithmetic, headless-Chrome gates), Lean 4 (statements, builds, axioms, two kernels), SAT, LaTeX; GPU campaigns on RunPod; Hosted Training.

Product: design systems and the visual grammar, Figma, Webflow (CMS, GSAP), API integrations, SEO, accessibility.

EXPERIENCE

cert-machine — Independent research engineer — 2026–present

The audits, the re-performance, the RL work and the grammar above; open source, one author.

Finsweet — Senior Webflow Developer — 2020–present

End-to-end delivery of Webflow sites for global clients, from concept to deployment: CMS architecture, custom logic, advanced animation.

Beacon / Embraer — Webflow Developer & API Integration Specialist — 2021–2023

Migrated complex WordPress systems to scalable Webflow stacks; built enterprise frontends integrated with internal APIs and third-party services.

Filmelier — Product Designer — 2019–2020

Editorial UX and a Figma-to-Webflow production pipeline; SEO and accessibility.

Brainn, Movile, ED Interactive, Make ID, TAG, LabCom — Designer — 2010–2019

Digital design, branding, UX/UI and product work.

EDUCATION

Universidade Estadual Paulista (UNESP) — Bachelor's in Industrial & Graphic Design — 2003–2009

SENAI São Paulo — BTech in Graphic and Printing Equipment Operation — 2004–2006